

SETUP POSTGRES LOCALLY

PostgreSQL is a **C-based relational database server**. When you "run Postgres locally," you're compiling C source code into a binary, initializing a data directory, and running a server process that listens on a TCP port (default: 5432).

Tool	Why needed
<code>gcc</code> / <code>clang</code>	Compiles C source → binary
<code>make</code>	Runs the build instructions (<code>Makefile</code>)
<code>git</code>	Clone the repo, manage branches
<code>pkg-config</code>	Helps <code>./configure</code> find libraries

```
xcode-select --install  
brew install pkg-config
```

Required Libraries

Library	Why needed
<code>libreadline-dev</code>	Interactive <code>psql</code> shell (arrow keys, history)
<code>zlib-dev</code>	Compression support (WAL, backups)
<code>flex</code>	Lexer generator — parses SQL syntax tokens
<code>bison</code>	Parser generator — builds the SQL grammar tree

```
brew install readline zlib flex bison
```

Optional but Recommended Libraries

Library	Why
<code>libssl-dev</code>	SSL connections between client & server
<code>libxml2-dev</code>	XML data type support
<code>libicu-dev</code>	Proper Unicode/collation support (needed for modern

Library	Why
	PG)
<code>python3-dev</code>	PL/Python procedural language
<code>libperl-dev</code>	PL/Perl procedural language

```
sudo apt install -y libssl-dev libxml2-dev libicu-dev
```

Getting the Source Code

```
git clone https://github.com/postgres/postgres.git
```

```
cd postgres
```

```
# List available branches/versions
git branch -a | grep -E 'REL_1[6-7]'
```

#Output:

```
adityadave@Adityas-MacBook-Air-3 postgres % git branch -a | g
rep -E 'REL_1[6-7]'
```

```
remotes/origin/REL_16_STABLE
remotes/origin/REL_17_STABLE
```

The Build Pipeline — Step by Step

Source Code

↓

```
./configure ← detects your system, libraries, sets pat
hs
```

↓

```
make ← compiles all C files → object files → bi
naries
```

↓

```
make install ← copies binaries to your chosen prefix di
```

```

r
  ↓
initdb          ← creates a fresh database cluster (data d
ir)
  ↓
pg_ctl start    ← starts the postgres server process
  ↓
psql           ← connect and run SQL

```

Set all env variables:

```

export ICU_PREFIX=$(brew --prefix icu4c)
export OPENSSL_PREFIX=$(brew --prefix openssl)
export READLINE_PREFIX=$(brew --prefix readline)
export LIBXML2_PREFIX=$(brew --prefix libxml2)

export PKG_CONFIG_PATH="\
$ICU_PREFIX/lib/pkgconfig:\
$OPENSSL_PREFIX/lib/pkgconfig:\
$LIBXML2_PREFIX/lib/pkgconfig:\
$READLINE_PREFIX/lib/pkgconfig"

export LDFLAGS="-L$ICU_PREFIX/lib -L$OPENSSL_PREFIX/lib -L$RE
ADLINE_PREFIX/lib -L$LIBXML2_PREFIX/lib"

export CPPFLAGS="-I$ICU_PREFIX/include -I$OPENSSL_PREFIX/incl
ude -I$READLINE_PREFIX/include -I$LIBXML2_PREFIX/include"

```

Configure:

```

./configure \
  --prefix=$HOME/pgsql \           # Install HERE (not system-wi
de, safe for dev)                  de)
  --enable-debug \                 # Include debug symbols (for
gdb, crash analysis)               gdb, crash analysis)
  --enable-cassert \              # Enable internal assertions

```

```
(catches bugs early)
--with-openssl \           # Enable SSL support
--with-libxml \           # Enable XML type
--with-icu                 # Enable ICU collations (required in PG16+)
```

```
./configure --prefix=$HOME/pgsql --enable-debug --enable-cassert --with-openssl --with-libxml --with-icu
```

Compile:

```
make -j$(sysctl -n hw.logicalcpu)
```

Install binaries:

```
make install
```

This copies everything to `$HOME/pgsql/`:

Build contrib modules: These are official extensions bundled with Postgres (e.g. `pg_stat_statements`, `uuid-oss`):

```
cd contrib
make -j$(sysctl -n hw.logicalcpu)
make install
cd ..
```

Set up your shell environment permanently:

```
cat >> ~/.zshrc << 'EOF'

# PostgreSQL dev build
export PGHOME=$HOME/pgsql
export PGDATA=$HOME/pgdata
export PATH=$PGHOME/bin:$PATH
export DYLD_LIBRARY_PATH=$PGHOME/lib # macOS equivalent of
```

```
LD_LIBRARY_PATH
EOF
```

```
source ~/.zshrc
```

verify:

```
adityadave@Adityas-MacBook-Air-3 postgres % which psql
/Users/adityadave/pgsql/bin/psql
```

Initialize the database cluster:

```
initdb -D $PGDATA
```

What this creates:

`$HOME/pgdata/`

— postgresql.conf	← main server config
— pg_hba.conf	← authentication rules
— base/	← your actual data will live here
— pg_wal/	← write-ahead log for crash recovery

Start the server:

```
pg_ctl -D $PGDATA -l $PGDATA/logfile start
```

Check it's running:

```
pg_ctl -D $PGDATA status
```

Connect!:

```
psql -d postgres
```

```
[adityadave@Adityas-MacBook-Air-3 postgres % psql -d postgres
psql (18.4)
Type "help" for help.

postgres=# \c
You are now connected to database "postgres" as user "adityadave".
postgres=# \l
```

Name	Owner	Encoding	Locale	Provider	Collate	Ctype	Access privileges
postgres	adityadave	UTF8	libc		en_IN.UTF-8	en_IN.UTF-8	
template0	adityadave	UTF8	libc		en_IN.UTF-8	en_IN.UTF-8	
template1	adityadave	UTF8	libc		en_IN.UTF-8	en_IN.UTF-8	

```
(3 rows)

postgres=#
```

```
[adityadave@Adityas-MacBook-Air-3 postgres % which psql
/Users/adityadave/pgsql/bin/psql
```

Done!

Now Start Debugging.

Restart Checklist:

- ☐ Open terminal
- ☐ `source ~/.zshrc` (only if psql not found)
- ☐ `pg_ctl -D $PGDATA start`
- ☐ `pg_ctl -D $PGDATA status`

Important commands:

```
# --- AFTER RESTART ---
source ~/.zshrc                                # reload env if ne
                                                # eded
pg_ctl -D $PGDATA start                        # start server
pg_ctl -D $PGDATA status                       # verify

# --- CODE CHANGE ---
make -j$(sysctl -n hw.logicalcpu)              # recompile
make install                                  # install
pg_ctl -D $PGDATA restart                     # apply
```

```
# --- CLEAN REBUILD ---
make clean
make -j$(sysctl -n hw.logicalcpu)
make install
pg_ctl -D $PGDATA restart

# --- SERVER CONTROL ---
pg_ctl -D $PGDATA start
pg_ctl -D $PGDATA stop
pg_ctl -D $PGDATA restart
pg_ctl -D $PGDATA status
tail -f $PGDATA/logfile # watch live logs
```